



SAPIENZA
UNIVERSITÀ DI ROMA

Master of Science in Engineering in Computer Science

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Human Computer Interaction Project Report

SapienzaAdvisor:

A Student-Centered Academic Platform

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Chapter 1 - Abstract

This project presents the development of SapienzaAdvisor, an innovative academic platform designed specifically for university students, with particular focus on those enrolled at Sapienza University of Rome. The system aims to enhance the academic experience by providing comprehensive tools for course evaluation, collaborative learning, and resource sharing within a personalized and structured environment.

The core functionality of SapienzaAdvisor revolves around enabling students to access detailed course information specific to their degree programs, contribute peer reviews and ratings, and share high-quality study materials with fellow students. The platform addresses the critical gap in structured academic resource sharing and course selection guidance that currently relies on scattered information across various informal channels.

Key stakeholders include undergraduate and graduate students, academic advisors, and faculty members seeking to understand course feedback and student needs. The system's primary use cases encompass informed course selection, collaborative note sharing, and peer-reviewed resource enhancement, all designed to streamline academic workflows while fostering a supportive and collaborative learning community.

Chapter 2 - Work Plan

Chapter 3 - Requirement Analysis

3.1 Competitor Analysis

An extensive analysis of existing platforms reveals significant gaps in the current academic support ecosystem. The competitive landscape includes several categories of platforms, each addressing only partial aspects of student needs:

Educational Management Systems such as Google Classroom and Moodle focus primarily on teacher-centric content delivery but lack student-driven evaluation and peer collaboration features. While these platforms excel in structured content distribution, they do not provide mechanisms for student feedback or collaborative resource sharing.

Study Resource Platforms like StuDocu offer document sharing capabilities but lack integration with specific degree programs and peer review systems. These platforms often contain unverified content without quality assurance mechanisms.

Academic Communication Tools including Piazza provide Q&A functionality but are limited in scope and do not offer comprehensive course evaluation or structured resource repositories.

Informal Communication Channels such as Discord, Telegram, and WhatsApp groups currently serve as primary information sharing venues but lack organization, searchability, and quality control.

Competitor Analysis

Feature	Sapienza Advisor	Google Classroom	Moodle	StuDocu	Piazza	Discord / Telegram	Google Drive
Degree Program Personalization	✓	✗	✗	✗	✗	✗	✗
Course Evaluation & Reviews	✓	✗	✗	✗	✗	✗	✗
Collaborative Note Sharing with Peer Review	✓	✗	✗	✓ (limited, no peer review)	✗	✗	✗
Integration with Academic Advisors & Faculty	✓	✓ (teacher-centric)	✓ (teacher-centric)	✗	✗	✗	✗
Structured Repository per Course	✓	✗	✓ (limited)	✗	✗	✗	✗
Community-Oriented Academic Support	✓	✗	✗	✗	✓ (Q&A only)	✓ (informal)	✗

SapienzaAdvisor differentiates itself through several killer features:

- **Degree Program Personalization:** Students register under their specific degree program, ensuring access to relevant courses and information
- **Comprehensive Course Evaluation System:** Multi-dimensional rating system covering teaching quality, difficulty, workload, and career relevance
- **Peer-Reviewed Resource Repository:** Quality-assured study materials with community-driven rating and review systems
- **Academic Integration:** Direct connection with academic advisors and faculty for enhanced support and feedback loops

3.2 User Profiles

Based on our project, we have identified one user profile that obviously needs to be a student.

Student User Profile:

- Age: 18-28 years
- Gender: Diverse representation across all genders
- Education Level: Bachelor's to PhD students
- Location: Primarily Rome-based, with consideration for international exchange students
- Technology Proficiency: High smartphone and computer literacy, native digital experience
- Academic Status: Active enrollment in university programs
- Motivation: Seeking efficient study methods, peer collaboration, and informed academic decision-making

3.3 Personas and Scenarios

3.3.1 Persona 1:

Donato Bianchi, 22 - Master's Student in Artificial Intelligence

Donato represents the methodical, ambitious student who values informed decision-making in academic planning. As a second-year master's student, he faces the critical task of selecting elective courses that will impact his career trajectory in AI research and development.

Background: Donato has just begun his Master's in Artificial Intelligence and is preparing his study plan for the upcoming semester. His undergraduate experience taught him the importance of careful course selection, but he finds the current information gathering process fragmented and unreliable.

Pain Points:

- Information scattered across WhatsApp groups, forums, and informal discussions
- Lack of structured comparison tools for courses
- Uncertainty about course difficulty and professor teaching styles
- No reliable source for peer experiences and recommendations

Scenario: Upon receiving the university's study plan submission deadline notice, Donato logs into SapienzaAdvisor. He navigates to his AI Master's program section and explores available elective courses. Using the platform's filtering system, he sorts courses by rating, workload, and career relevance.

Reading detailed peer reviews, Donato discovers that the "Advanced Machine Learning" course receives high ratings for practical applications and clear instruction methodology. The reviews

highlight specific aspects like project-based learning and industry connections, information that was previously inaccessible through formal channels.

He downloads highly-rated study materials from previous students, including course summaries and project examples, giving him confidence in his selection. The platform's structured approach allows him to make informed decisions and submit his study plan with conviction.

3.3.2 Persona 2:

Sofia Romano, 21 - Biomedical Engineering Student

Sofia embodies the dedicated student struggling with information overload and time management challenges. As a third-year student facing increasingly complex coursework, she requires efficient access to high-quality study resources.

Background: Sofia is passionate about biomedical engineering but often feels overwhelmed by the volume of material she needs to master. Her study sessions are frequently inefficient due to time spent searching for reliable resources rather than actual learning.

Challenges:

- Difficulty finding well-organized, high-quality study materials
- Time wasted on scattered, incomplete resources
- Lack of peer connection for study group formation
- Uncertainty about material quality and relevance

Scenario: Three weeks before her Biomechanics exam, Sofia struggles with complex mathematical proofs and theoretical concepts. Previous attempts to find study materials through social media groups yielded fragmented, unreliable information.

Accessing SapienzaAdvisor, Sofia filters available materials by course, rating, and content type. She quickly identifies highly-rated summary notes that received positive peer reviews for clarity and comprehensiveness. The platform's quality assurance system, based on student ratings and academic verification, ensures she accesses reliable content.

Additionally, Sofia finds supplementary exercise sets with detailed solutions, uploaded by students who successfully completed the course in previous years. The platform's review system helps her identify the most effective study strategies and connect with peer study groups preparing for the same examination.

3.3.3 Persona 3:

David Müller, 23 - Exchange Student from Germany

David represents the international student demographic, facing unique challenges related to cultural adaptation, language barriers, and unfamiliar academic systems.

Background: David is participating in an Erasmus exchange program, studying Economics at Sapienza for one semester. Despite his excitement about the international experience, he faces significant challenges in navigating the Italian university system and academic culture.

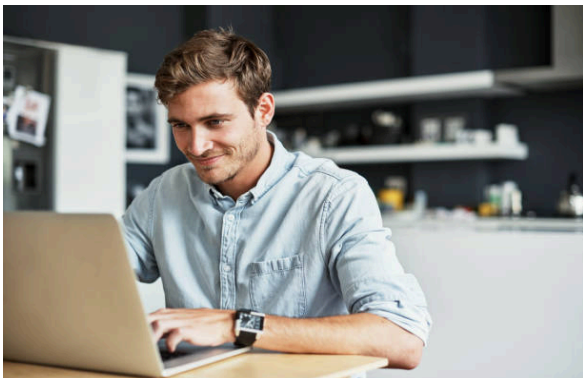
Specific Challenges:

- Unfamiliarity with Italian academic system and procedures
- Language barriers in accessing course information
- Uncertainty about course equivalencies for his home university
- Limited local network for academic guidance

Scenario: Two weeks into his exchange program, David needs to finalize his course selection while ensuring compatibility with his home university requirements. Language barriers make it difficult to fully understand course descriptions and academic expectations.

Using SapienzaAdvisor's international student features, David filters courses by language of instruction and accesses English-language reviews and materials. The platform's detailed course descriptions, enhanced by peer reviews, provide clarity about expectations, assessment methods, and workload.

Through the platform, David connects with other international students and finds study materials translated or created in English. The community aspect of SapienzaAdvisor helps him integrate into the academic environment while maintaining academic success aligned with his home university requirements.



Donato Bianchi



Sofia Romano



David Müller

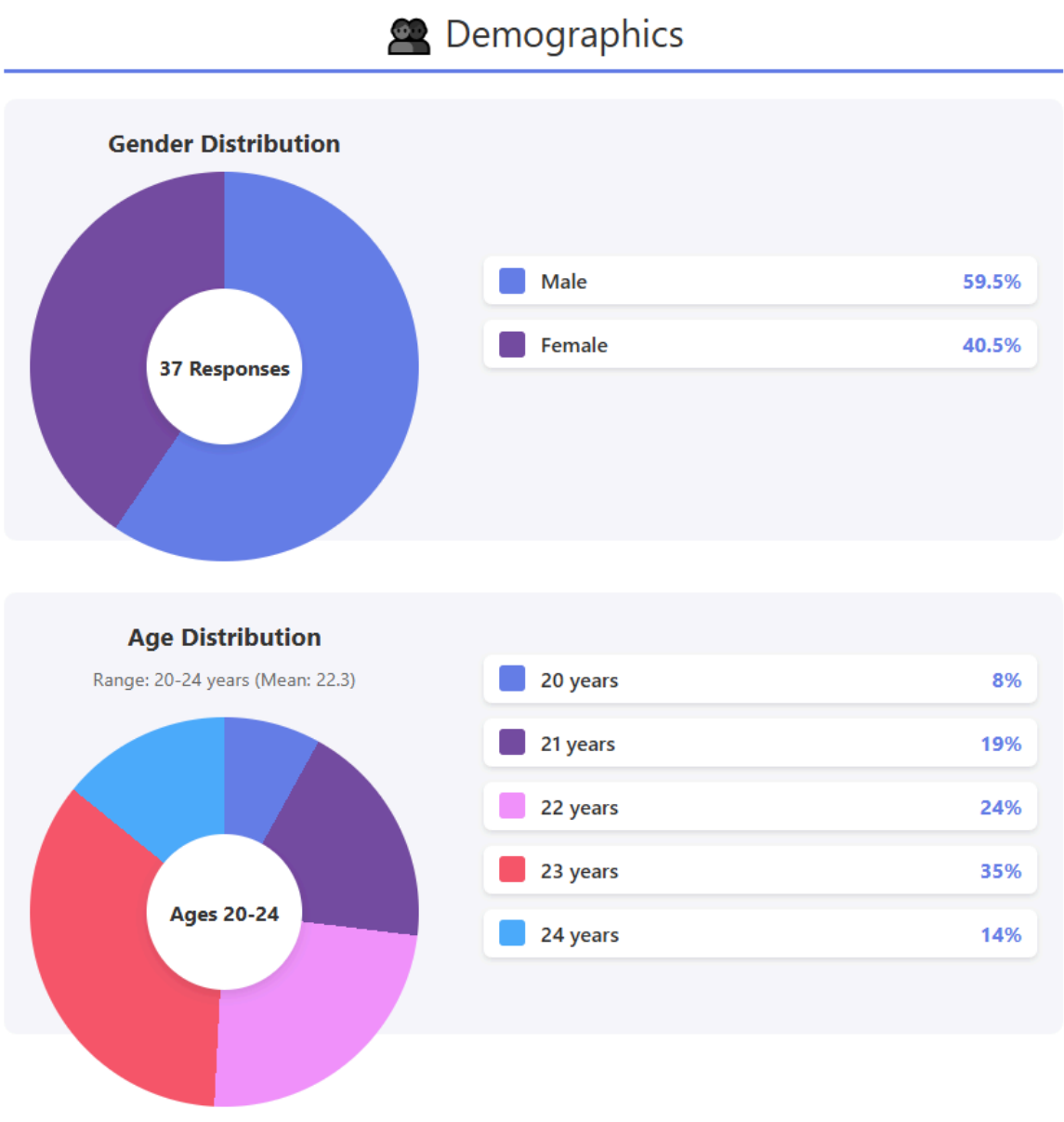
3.4 Questionnaire Analysis

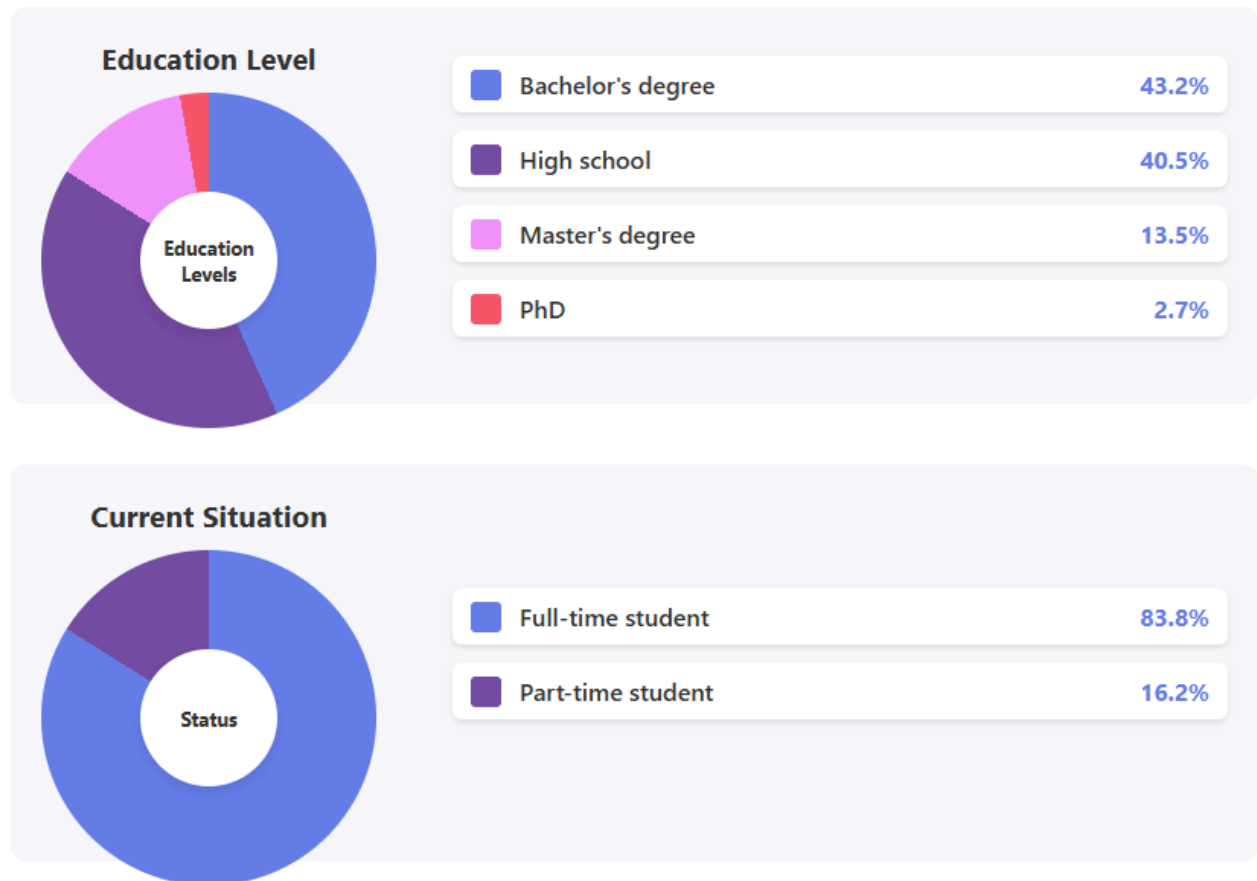
To validate our user requirements and understand specific needs, we conducted comprehensive questionnaire research targeting our three primary user groups. The data collected provides crucial insights into user behaviours, preferences, and pain points.

3.4.1 Questionnaire Results Summary

Our comprehensive questionnaire received **37 responses** from students across various university programs. The demographic analysis reveals a representative sample of our target user base:

Demographics Overview:



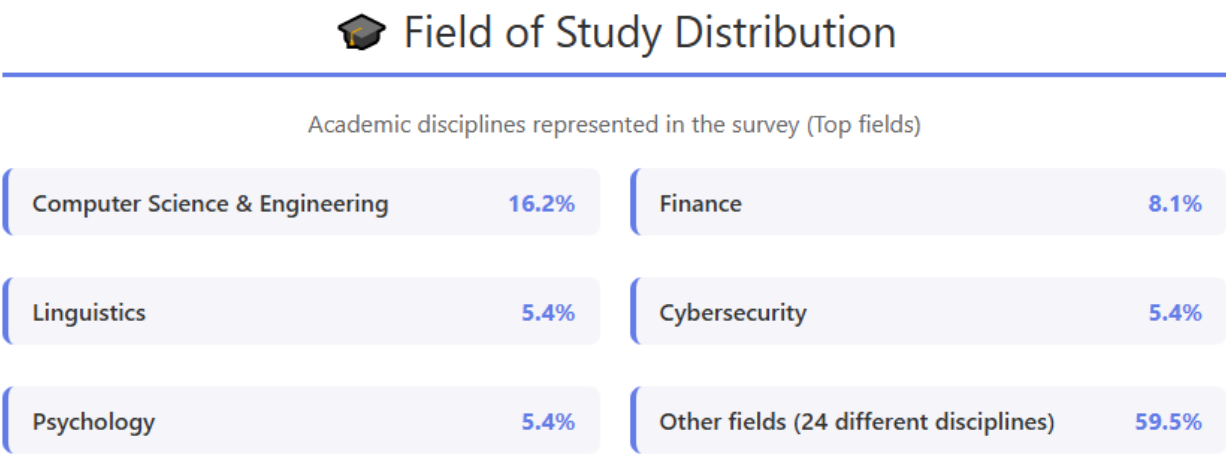


The age distribution centers around the traditional university range of 20-24 years (mean: 22.3), with the majority being 23 years old (35%), followed by 22 years old (24%). This concentration reflects students in their peak undergraduate and early graduate years, when course selection decisions are most critical.

Gender representation shows 59.5% male and 40.5% female participants, providing balanced perspectives across gender lines. Education levels reveal a mix of academic stages: 43.2% hold Bachelor's degrees, 40.5% are completing high school (likely incoming students), 13.5% have Master's degrees, and 2.7% are pursuing PhDs. This distribution captures students across different decision-making stages, from initial university enrollment to advanced academic planning.

Importantly, 83.8% are full-time students, confirming our primary target demographic's availability and engagement level. The remaining 16.2% represent part-time students with varying work commitments, highlighting the need for flexible platform features that accommodate different time constraints and usage patterns.

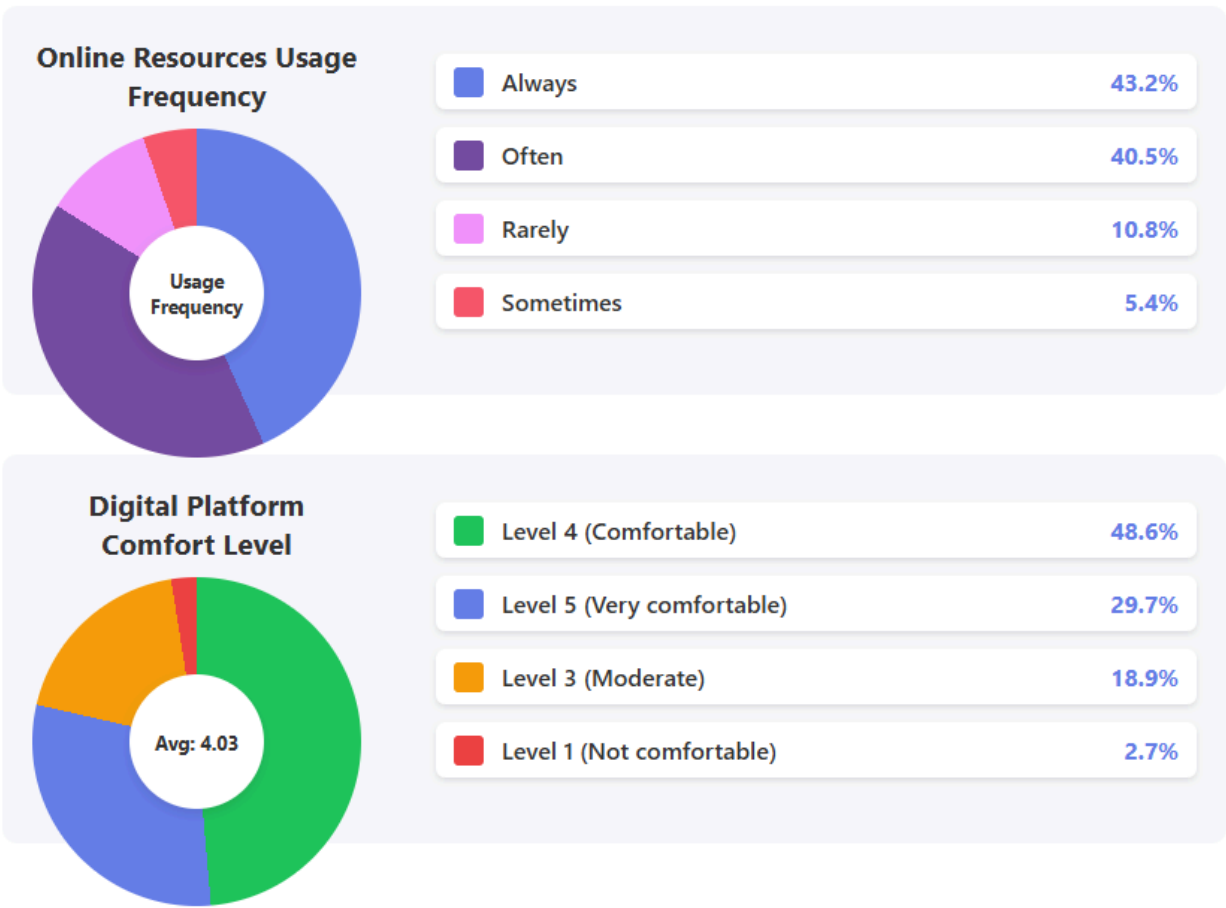
Field of Study Distribution:



The survey captured responses from 27+ different academic disciplines, with Computer Science & Engineering leading at 16.2%, followed by Finance (8.1%). This diversity validates the platform's broad market potential across various academic fields and confirms the need for a flexible system that can accommodate different disciplinary requirements and academic cultures.

Digital Habits and Comfort:

 Digital Habits & Comfort

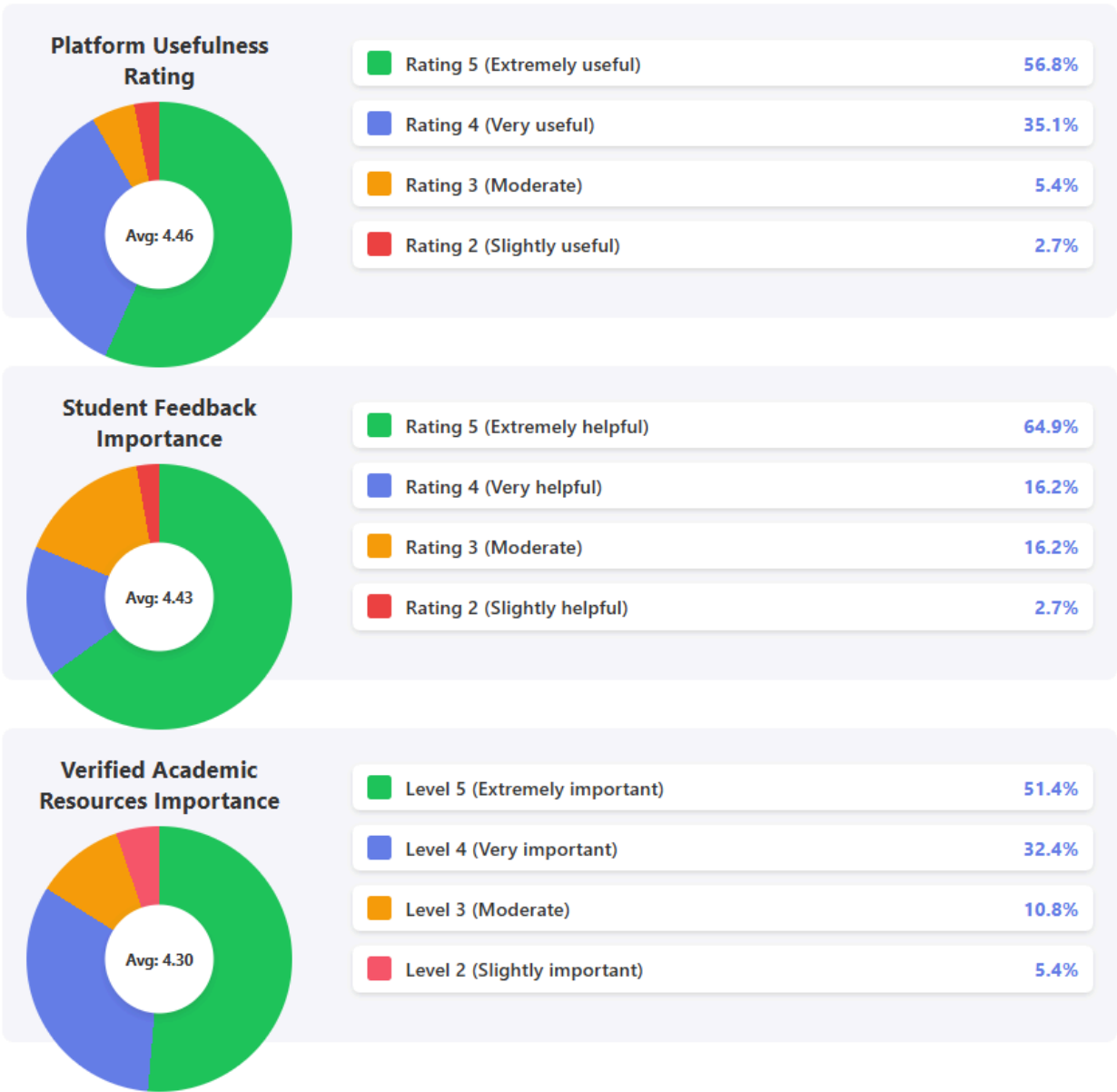


Online Resources Usage Frequency: The data shows strong digital adoption among students, with 83.7% using online study resources either "always" (43.2%) or "often" (40.5%). This high usage frequency validates the digital-first approach of SapienzaAdvisor and confirms that students are already comfortable with accessing academic materials through digital platforms. Only 10.8% rarely use online resources, indicating minimal resistance to digital solutions.

Digital Platform Comfort Level: Students demonstrate high comfort with digital platforms, averaging 4.03/5 on the comfort scale. A significant 78.3% rated their comfort level as 4-5, with 48.6% at level 4 (comfortable) and 29.7% at level 5 (very comfortable). This digital fluency supports the feasibility of implementing a comprehensive online academic platform.

Platform Value Assessment:

★ Platform Value Assessment



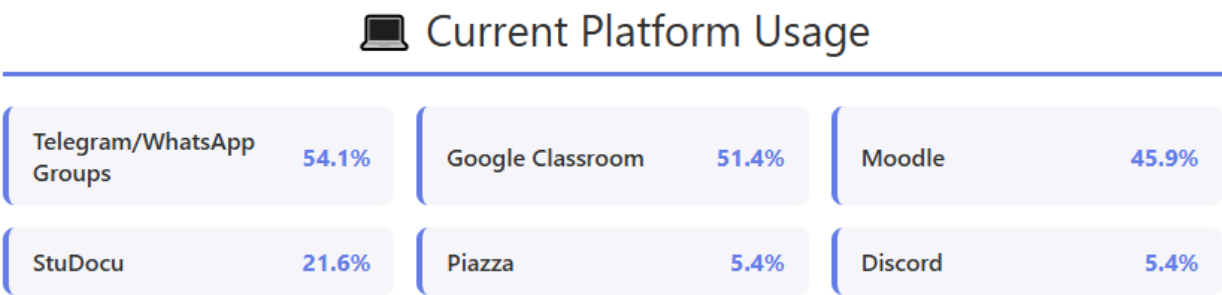
Platform Usefulness Validation: The data strongly validates our project concept, with an overwhelming 92% of respondents rating the platform usefulness as 4 or 5 on a 5-point scale (average: 4.46/5). This indicates extremely high demand for a centralized academic platform.

Student Feedback Value: 81% of students rated peer feedback as highly valuable (4-5 on scale), with an average rating of 4.43/5, confirming the importance of our peer-review system.

Verified Academic Resources Importance: Students place high value on quality assurance, with 83.8% rating verified academic resources as important (levels 4-5), averaging 4.30/5. This strong

preference for verified content validates our emphasis on peer review systems and academic verification mechanisms. The high importance rating (51.4% gave the maximum rating) indicates that quality control features will be crucial for platform adoption and trust-building.

Current Platform Landscape:



The questionnaire reveals a highly fragmented digital ecosystem where students must navigate multiple platforms to gather academic information. The dominance of informal communication channels is striking: Telegram/WhatsApp Groups lead usage at 54.1%, highlighting students' reliance on unstructured, peer-to-peer information sharing despite its inherent limitations in searchability and quality control.

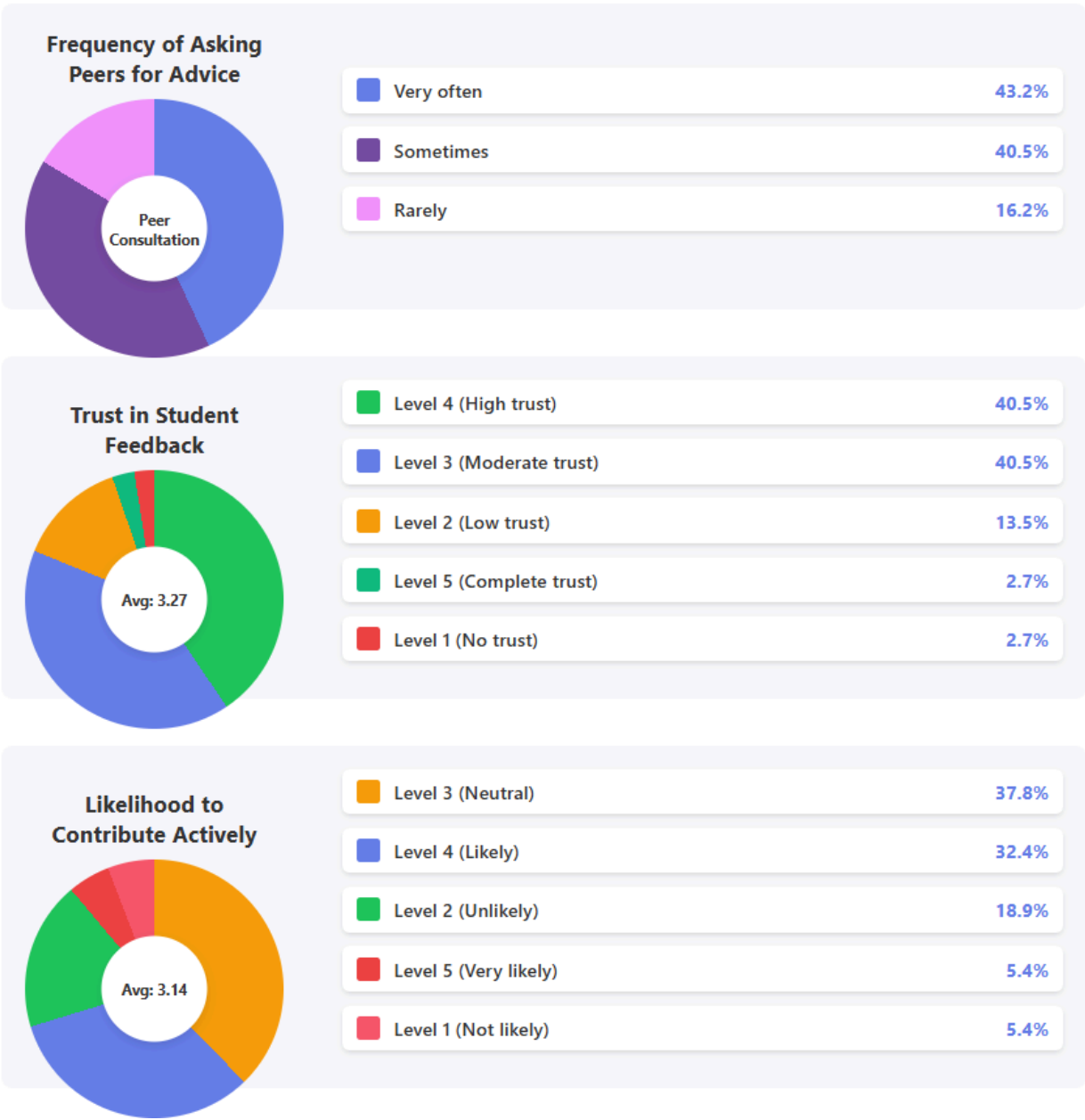
Institutional platforms show significant but incomplete adoption: Google Classroom (51.4%) and Moodle (45.9%) serve primarily as content delivery systems rather than collaborative learning environments. These teacher-centric platforms lack the peer evaluation and community-driven features that students actively seek.

External platforms like StuDocu (21.6%) demonstrate student willingness to seek alternatives for study materials, though these solutions often lack integration with specific university programs and quality verification mechanisms. The minimal usage of specialized academic discussion platforms like Piazza (5.4%) and Discord (5.4%) suggests that current solutions fail to meet students' comprehensive academic support needs.

This fragmentation forces students to maintain multiple accounts, cross-reference information across platforms, and rely on informal networks for critical academic decisions, creating inefficiencies that SapienzaAdvisor's unified approach directly addresses.

Social Behaviour and Trust:

📍 Social Behavior & Trust



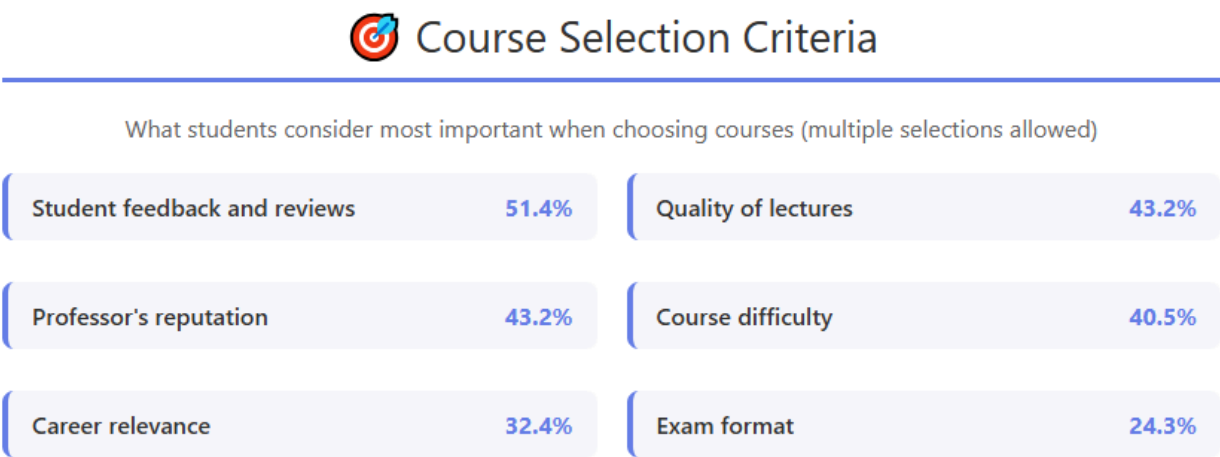
Frequency of Seeking Peer Advice: The data reveals a strong peer consultation culture, with 83.7% of students asking peers for academic advice either "very often" (43.2%) or "sometimes" (40.5%). This behavior validates the social learning approach underlying SapienzaAdvisor and confirms that students naturally turn to their peers for academic guidance, supporting the platform's community-driven model.

Trust in Peer Feedback: The data reveals moderate but significant trust in student feedback (average: 3.27/5), with 83% of students showing neutral to positive trust levels. This validates the need for quality assurance mechanisms in our platform.

Likelihood to Contribute Actively: Student willingness to contribute content shows moderate levels, averaging 3.14/5. While 70.2% show some willingness to contribute (levels 3-5), with 32.4% likely to contribute (level 4), the moderate average suggests the need for incentive mechanisms to encourage active participation. This finding informs the design of engagement strategies to ensure sustainable content generation.

Course Selection Criteria Analysis:

Students prioritize multiple factors when selecting courses, indicating the need for comprehensive information:

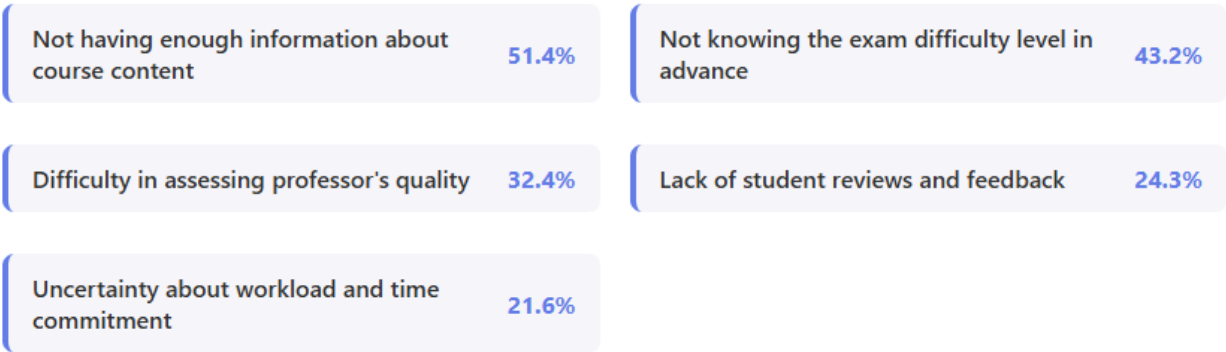


Notably, "Student feedback and reviews" ranks as the top criterion (51.4%), directly validating SapienzaAdvisor's core value proposition. This preference, combined with "Quality of lectures" (43.2%) and "Professor's reputation" (43.2%), demonstrates that students seek comprehensive, peer-validated information about courses, which traditional systems fail to provide systematically.

Main Pain Points Identified:

! Main Difficulties in Course Selection

Primary challenges students face when choosing university courses

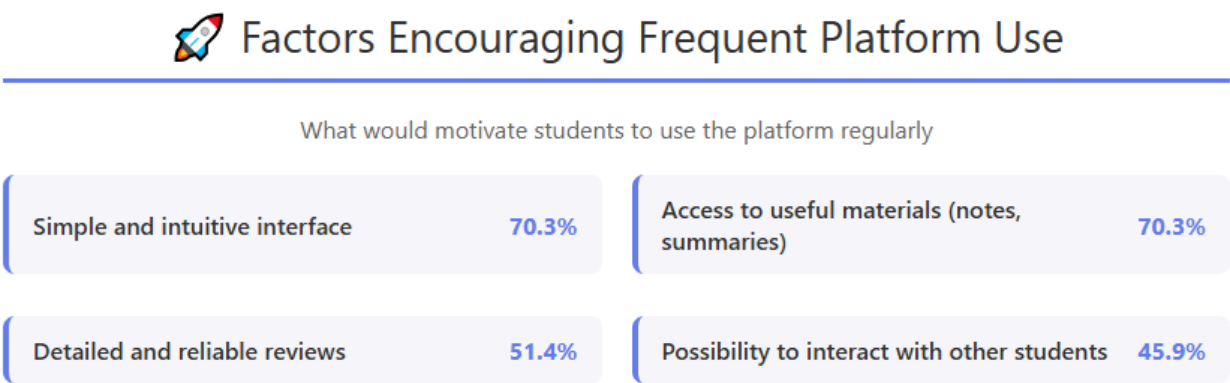


The questionnaire reveals significant barriers in current course selection processes that directly validate SapienzaAdvisor's value proposition. The primary challenge, affecting 51.4% of students, is "not having enough information about course content," indicating a fundamental gap between what universities provide and what students need for informed decision-making.

The second major pain point, "not knowing the exam difficulty level in advance" (43.2%), highlights students' desire for practical, peer-validated information about academic expectations. This uncertainty forces students to make critical academic decisions without understanding the real workload and assessment challenges they will face.

"Difficulty in assessing professor's quality" ranks third (32.4%), demonstrating that official course descriptions fail to capture the teaching effectiveness and instructor characteristics that significantly impact student success. Additional challenges include "lack of student reviews and feedback" (24.3%) and "uncertainty about workload and time commitment" (21.6%).

Encouraging Factors:



The top factors encouraging frequent platform use are "Simple and intuitive interface" and "Access to useful materials" (both at 70.3%), followed by "Detailed and reliable reviews" (51.4%). This data directly informs UX design priorities, emphasizing the need for streamlined interface design and ensuring the platform provides genuine value through high-quality content and comprehensive reviews.

2.4.2 Design Implications

The questionnaire findings provide a compelling roadmap for SapienzaAdvisor's development strategy and feature prioritization. The overwhelming validation of platform usefulness (92% rating 4-5/5) not only confirms market demand but also suggests strong user willingness to adopt a comprehensive solution that addresses current pain points.

The clear alignment between student needs and our proposed features creates a natural development pathway. With 73% of students prioritizing material sharing and 67.6% seeking detailed course reviews, our core functionality directly matches user expectations. This alignment reduces adoption friction and provides clear metrics for measuring platform success.

However, the moderate trust levels in peer feedback (3.27/5 average) highlight a critical design challenge that transforms into an opportunity. By implementing robust quality assurance mechanisms, verified review systems, and transparent contribution tracking, SapienzaAdvisor can differentiate itself from existing informal solutions while building the credibility necessary for widespread adoption.

The current platform fragmentation presents a strategic advantage rather than a competitive threat. Students already demonstrate comfort with digital tools (4.03/5 average) and actively seek peer consultation (83.7%), indicating readiness for a unified solution. Our challenge lies not in changing user behavior but in consolidating existing practices into a more efficient, reliable system.

The emphasis on interface simplicity (70.3% prioritize intuitive design) reinforces the importance of user experience over feature complexity. This finding suggests that success depends on seamless integration of powerful functionality within an accessible, streamlined interface that reduces rather than increases cognitive load for users already managing multiple platforms.

Chapter 3 - Task Analysis: HTA & STN

Following comprehensive requirement analysis, we developed detailed Hierarchical Task Analysis (HTA) and State Transition Networks (STN) for our platform's core functionalities. These analyses provide structured understanding of user workflows and inform interface design decisions.

3.1 Find and Select Course Reviews

Goal: Find and select a useful review for a university course

Main Task Breakdown:

0. Find and select a useful review for a university course

- **1. Access the platform**
- **2. Search for a specific course**
 - 2.1 Navigate to faculty section
 - 2.2 Use course name search functionality
 - 2.3 Select desired course from results
- **3. Examine course reviews**
 - 3.1 View review list with preview information (rating, date, keywords)
 - 3.2 Sort reviews by relevance, date, or rating
 - 3.3 Filter reviews using metadata (professor, academic year, difficulty level)
 - 3.4 Read complete review content
- **4. Evaluate usefulness of found reviews**
 - 4.1 Check community helpfulness votes
 - 4.2 Verify attached study materials or resource links
- **5. Select review to inform academic decisions**

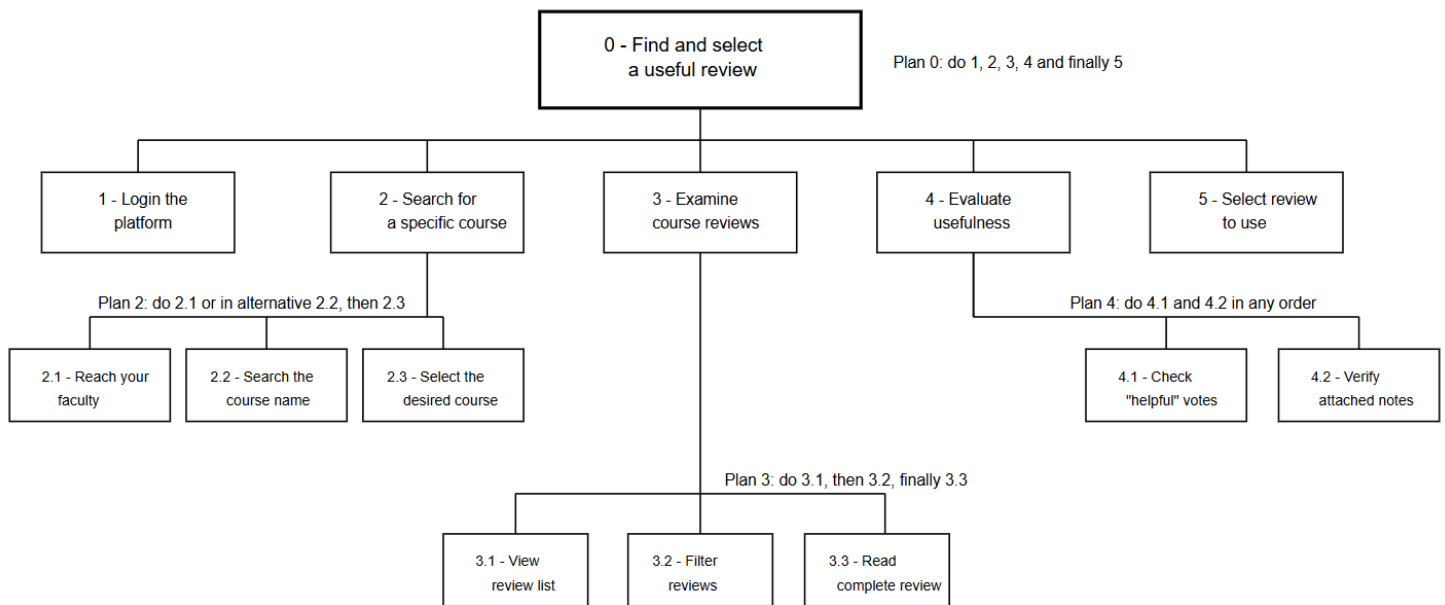
Execution Plans:

- **Plan 0:** Execute tasks 1 → 2 → 3 → 4 → 5 in sequence
- **Plan 2:** Execute 2.1 OR 2.2, then proceed to 2.3
- **Plan 3:** Execute 3.1 → 3.2 → 3.3 in sequence for comprehensive review examination
- **Plan 4:** Execute 4.1 AND 4.2 in any order for thorough evaluation

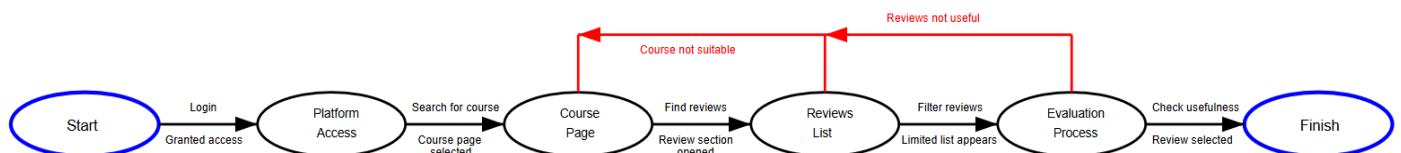
Key Design Considerations:

- Multiple search pathways accommodate different user preferences
- Progressive information disclosure prevents cognitive overload
- Community validation mechanisms build trust and reliability
- Flexible filtering system supports diverse user needs

HTA for "Find and select a useful review for a university course"



STN for "Find and select a useful review for a university course"



3.2 Share Study Materials

Goal: Add study material to the platform

Main Task Breakdown:

0. Add study material

- **1. Access the platform**
 - 1.1 Open the webpage
 - 1.2 Complete login process
- **2. Prepare the material**
 - 2.1 Choose material type (notes, summaries, exercises, etc.)
 - 2.2 Check content format compatibility
 - 2.3 Ensure quality standards
- **3. Select target course**
 - 3.1 Search course database
 - 3.2 Choose correct course and section
- **4. Upload process**
 - 4.1 Add descriptive title
 - 4.2 Write comprehensive description
 - 4.3 Add relevant keywords or tags
 - 4.4 Attach material files
- **5. Publish content**
 - 5.1 Save as draft
 - 5.2 Confirm and publish publicly
- **6. Monitor and manage**
 - 6.1 Read community reviews and feedback
 - 6.2 Edit or delete content as needed

Execution Plans:

- **Plan 0:** Execute tasks 1, 2, 3, 4, 5 in order
- **Plan 1:** Execute 1.1, then 1.2 sequentially
- **Plan 2:** Execute 2.1, 2.2, 2.3 in any order; if quality check fails, repeat relevant steps
- **Plan 3:** Execute 3.1 then 3.2; if course not found, repeat 3.1 with different criteria
- **Plan 4:** Execute 4.1, 4.2, 4.3, 4.4 in any order; if upload fails, check format and repeat 4.4
- **Plan 5:** Execute 5.1, then 5.2 in order
- **Plan 6:** Execute 6.1; if issues identified, optionally execute 6.2

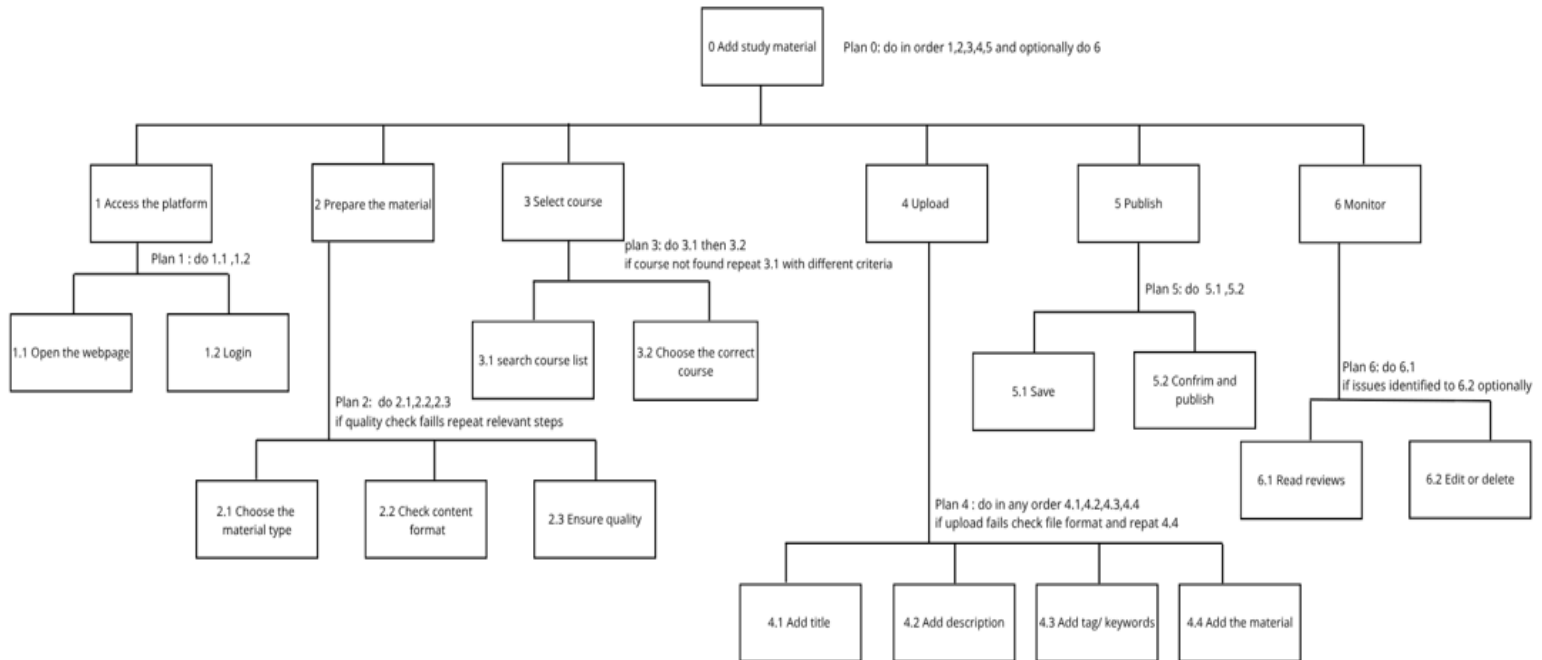
Quality Assurance Considerations:

- Material preparation phase ensures content quality before upload
- Course selection verification prevents misclassification
- Draft functionality allows review before publication
- Community feedback loop maintains ongoing quality standards

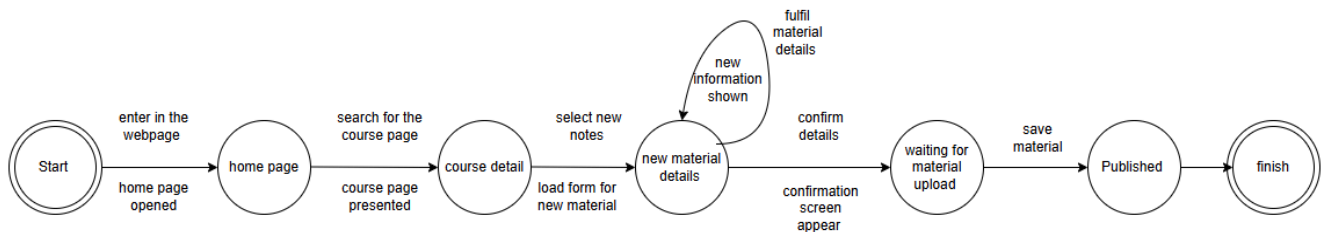
Error Handling:

- Format compatibility checks prevent upload failures
- Course verification reduces content misplacement
- Quality checkpoints maintain platform standards
- Edit/delete functionality allows content management

HTA for “Add study material to the platform”



STN for “ Add study material to the platform”



Chapter 5 - Evaluation Technique